PHASE 2 SPECIFICATION

Requirements Analysis & System Specification

Project: FitTrack AI – Intelligent Fitness Tracking & Health Insights Platform

1. FUNCTIONAL REQUIREMENTS

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- **Authentication & User Authorization:** Support user registration, login, and profile management. Enforce password hashing using `bcrypt` and protect private routes using JSON Web Tokens (JWT).
- **Activity & Fitness Management:** Allow authenticated users to perform full CRUD operations (add, view, update, delete) on fitness activities and user logs stored in MongoDB.
- **AI-Driven Health Insights:** Generate personalized fitness recommendations, workout guidance, and health summaries using AI integration based on user profile and activity history.
- **Reliability & Rule-Based Fallback:** Implement robust local rule-based fallback mechanisms to ensure uninterrupted system availability and fallback responses when AI services or external APIs are unreachable.

2. NON-FUNCTIONAL REQUIREMENTS

- **Security:** Secure user authentication using JWT authorization tokens, password encryption via `bcrypt`, CORS policy configuration, and strict API input validation.
- **Availability & Resilience:** Robust error handling and fallback logic to handle external service outages smoothly without interrupting end-user workflows.
- **Scalability & Hardware Specifications:** A modular MVC architecture with Node.js and MongoDB. Recommended environment: minimum 8 GB RAM and 1 GB storage. Configuration managed securely via environment variables.

3. TECHNICAL STACK & DEPENDENCIES

- **Runtime & Framework:** Node.js, Express.js (Modular MVC-style RESTful Backend)
- **Database:** MongoDB (User accounts, activity records, and fitness data)
- **Authentication:** `jsonwebtoken` (JWT), `bcrypt` / `bcryptjs`
- **AI & Rule Engine:** Integrated AI Health Insights API with Rule-Based Fallback logic

4. ACCESS CONTROL MATRIX

ENDPOINT TYPE	ACCESSIBILITY	CAPABILITIES SUPPORTED
Public API	Anyone	Access basic public endpoints and system status.
Protected API	Authenticated Users (JWT)	Profile management, workout & activity logging (CRUD), and receiving personalized AI health insights.